Switching Gene Expression On and Off

Biology 2e chapter 16 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

Selection and expression answer different questions. A selective plate tests growth under its conditions, while a reporter shows detectable gene activity. Compare a nutrient plate, an antibiotic control, and reporter-bearing cells with and without an inducer.

Investigation question

How can cells carry a reporter gene without visibly expressing it?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Use prepared cells in this model; it does not simulate transformation efficiency. Incubate each plate, then switch on the fluorescence viewer and record colony count plus brightness.

Select the indicated tool in the equipment tray, then select its target on the apparatus. Follow the preparation checklist and observe the changes. Complete three spreader passes before moving the closed plate to incubation. Observe growth and fluorescence separately. Review observations, then click Collect data and continue. Complete the assigned comparison before exploring.

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Biology 2e chapter 16 reading connection | Student worksheet / observations

Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: visible colonies

Condition / sample	Repeat	Source	Colonies	Fluorescence 0/1
No plasmid / nutrient agar	1			
No plasmid / antibiotic	1			
Reporter plasmid / antibiotic	1			
Reporter plasmid / antibiotic + arabinose	1			

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Biology 2e chapter 16 reading connection | Student worksheet / detailed record

Copy this page for each trial. Treatment: _____ Repeat: ____ Source: _____

hours	visible colonies	Notes
0		
6		
12		
18		
24		

Graph your observations

Label axes and units; keep each treatment and source identifiable. For repeated measurements, connect readings only from the same sample. For ordinal scores, explain the categories instead of treating the scale as concentration.

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Biology 2e chapter 16 reading connection | Student worksheet / interpretation

Which comparison tests antibiotic selection and which tests induction?

Does growth without fluorescence prove that the reporter gene is absent?

How would an incorrect plate label change your interpretation?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Generic arabinose-regulated reporter, not the lac operon. Colony appearance is a simplified visibility model, not bacterial generation time. Supplied prepared cells omit transformation; a single plate per condition cannot estimate variability.